Conversation as data: Extracting features from naturalistic conversation datasets

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Introduction

Every concept in social psychology plays out in conversation. It is how we seek advice, process emotions, build rapport, share information, learn new skills, and check in to see if our own understanding of the world aligns with others. Conversation is how truth, gossip, and misinformation spreads. It is how people coordinate actions and create systems that can connect us more tightly or divide us further. More frequent and higher quality conversations are a buffer against mental and physical health risks (Holt-Lunstad et al., 2015). Put simply, our conversations impact almost every part of our lives.

How does this fundamental human behavior shape our health, emotions, relationships, and decisions? Despite its central role in our lives, we understand surprisingly little about how conversation works. A major reason for this gap is methodological. A common practice in social psychology is to study complex phenomena by isolating one element at a time. However, real conversation is a multi-channel experience. People communicate through body language, facial expressions, tone of voice, and words—dynamically and all at once (Holler & Levinson, 2019). Trying to isolate a single variable in this context leads to interactions that are artificial and stilted, missing the essence of everyday conversations. Being able to study real conversations directly, reliably, and at scale is an important step towards understanding this most ubiquitous social behavior.

Clarifying precisely how conversation works—including the conditions under which its dynamics shift and outcomes change—will help reveal what makes conversation meaningful and effective. Identifying these key features may enable new strategies to amplify the positive effects of conversation. Such insights could also reveal what makes this ancient technology of face-to-face communication so uniquely powerful—and what might be missing in newer forms of

interaction. As people increasingly shift from in-person interactions to computer-mediated exchanges (Baym et al., 2004; Okdie et al., 2011) and even from human dialogues to conversations with chatbots (Brandtzaeg & Følstad, 2017), understanding the mechanics of conversation takes on new urgency. A deeper grasp of how conversation works can inform interventions to enhance its benefits while helping us critically evaluate how these emerging forms of communication compare—and what they might lack.

In this chapter, we outline key considerations for designing and analyzing conversation studies. We advocate for the use of representative designs to ensure that datasets have sufficient variation for statistical inference and to adequately reflect the complexity inherent in conversation. We then focus on how these conversation recordings can be quantified and suggest principles to assess the validity of these measurements to aid in answering psychological questions. Finally, we review analytical considerations when working with conversation data that often violates assumptions in parametric tests. Understanding how to work with natural conversation data carefully and rigorously will lead to robust results that the entire field can build upon.

How representative study designs enable better science

In order to understand how conversation truly works, we need to study conversations that are representative of those in the "real world" (Brunswik, 1955). *Representative designs* emphasize studying behavior in contexts that reflect the range of experiences people encounter in everyday life. Rather than tightly controlling variables to minimize "noise", this approach treats natural variation as an informative signal—while simultaneously minimizing sources of error. Increasing the variability of the psychological construct of interest ensures that the complexity of

the signals can be reliably modeled (Nastase et al., 2020; Westfall et al., 2016; Yarkoni, 2020), allowing for more ecologically valid insights.

Studies with representative designs tend to generalize better to real-world contexts, rather than overfitting our understanding to overly-constrained laboratory environments (Jolly & Chang, 2019; Lockman & Yu, 2024). The field of machine learning has already benefited from this insight and has largely moved away from building models overfit to toy problems towards training models on large, "noisy" datasets. These models generalize well to novel contexts because the distribution of the novel data was already approximated by the highly-variant training data.

Extending this idea further highlights the importance of collecting, curating, and sharing large, high-quality datasets of many different kinds of natural conversations—spanning diverse cultures, goals, relationship types, group sizes, and communication environments (J. P. Chang et al., 2020; Yeomans et al., 2023). The work of amassing these conversation datasets is a foundational step toward a more generalizable science of conversation.

The importance of reviewing conversation recordings

Once you have a conversation dataset, it will be tempting to immediately transform the conversation recordings into numeric values that lend themselves to quantitative analysis. It is important to resist that urge and instead first familiarize yourself with the data. Most of the conversations we experience are ones we actively participate in, which limits our ability to form a priori hypotheses. Our intuitions are guided by our own narrow experiences and idiosyncratic preferences, making them inherently biased. By reviewing a representative sample of conversation recordings, insights can emerge bottom-up from the data itself, revealing unexpected patterns and dynamics (Schroer & Yu, 2024). This kind of careful qualitative work is

foundational in the field of Conversation Analysis (Liddicoat, 2021; Stivers & Rossi, 2025), where close inspection of individual interactions has led to important insights that inspire new lines of research.

These qualitative reviews should occur throughout a project's lifecycle to ensure alignment between the research goals and the analytic choices (de Barbaro et al., 2013). Every step in an analysis pipeline will bring you further and further away from the conversation itself, with each new transformation adding another layer of abstraction. This process is valuable, but only insofar as it remains faithful to the underlying data. By analogy, imagine swimming in the ocean while keeping an eye on your beach umbrella, anchored in the sand. As the current pulls you farther out, periodically locating the umbrella allows you to gauge how far you've drifted—and to swim back when necessary. In the same way, regularly returning to the original recordings helps ensure that analyses stay grounded in the conversations themselves. Ultimately, to understand what happens in a conversation, we need to look at the conversation itself.

Feature extraction: Quantifying conversation behavior

There is no better source of information about what happens in a conversation than a recording of the conversation itself. Instead of relying on survey questions about what people *think* they do (Paulhus & Vazire, 2007), recordings allow us to analyze what people *actually* do. A conversation recording—video, audio, or transcript—is a rich source of data that can be quantified and examined. New computational tools make it possible to measure conversational behavior reliably and at scale—a process known as *feature extraction*.

Feature extraction tools often use *machine learning*, a broad set of computational methods designed to identify structure in data. Most machine learning models undergo a training

phase, during which they learn the statistical regularities that distinguish one category or pattern from another in a large collection of examples.

In *supervised learning*, models are trained on data with known labels. This allows the models to find complex nonlinear structure in a set of features to predict a set of predefined labels. For example, a vision model might be shown thousands of images labeled as "dog" or "cat." Through exposure to these examples, the model estimates which visual features represented from image pixels—such as shape, texture, or color—best predict each label. Once trained, it can classify new, unlabeled photos of dogs and cats with high accuracy. In conversation research, similar models are used to detect smiles, identify gestures, or classify the emotional valence of language.

Unsupervised learning does not rely on labels and instead learns structure directly from the data. Self-supervised learning, a type of unsupervised learning, is a powerful technique to train models based on discriminating real data from augmented data, predicting missing data, or providing simple rules such as contrastive learning where some data should be more similar relative to other data. These techniques allow researchers to estimate their model parameters without having to explicitly label the data. Large language models use this kind of approach—they learn from massive collections of text which words tend to co-occur, allowing them to predict what word is likely to come next or to generate coherent sentences.

In this chapter, our focus is not on how to train new models from scratch, but rather how to use and evaluate existing ones. Here, we will discuss the kinds of features that can be extracted from conversation data (Fig 1). We will include some citations to specific models as examples, but note that these models are ever-changing and others may be better suited for your particular dataset. Many pretrained models are publicly available through repositories like

Hugging Face or Github. These models can be used "off the shelf," modified, or combined depending on the research question. Having a sense of how a model was trained, what it was designed to do, and how well it performs can help you decide if it is the correct model for your use case.

Turn-taking

People take turns speaking and listening in conversation (Sacks et al., 1974). By quantifying when people are speaking or not, it is possible to compute metrics such as *total speaking time* (useful for determining whether one person dominated the conversation or if speaking time was evenly shared between all speakers), *average turn length* (which gives a sense of the pace of a conversation—does this conversation have lots of back-and-forths or do people tend to hold the floor for longer moments at a time), and *average gap length* (which gives a sense for how quickly people respond to each other, including the amount of overlapping speech and long pauses).

Defining what a psychologically meaningful "turn" is in a conversation can be a deceptively tricky question (Cooney & Reece, 2025). Models that define turn transitions purely from audio information alone may yield different sets of turns from models that incorporate semantic information into those decisions (Lin et al., 2025; Razavi et al., 2019). This distinction is most salient during moments of overlapping speech—frequent in natural conversation—where decisions need to be made about whether turns get broken up, or kept together. Including timestamps for the starts and ends of turns can keep information about overlaps intact. It is also possible to get word-level timestamp information, allowing researchers to define different sets of turn-taking rules depending on the research question.

The meaning of these turn-taking dynamics can vary with context (E. M. Templeton & Wheatley, 2023). For example, overlap in conversation can indicate increased engagement and mental alignment (E. M. Templeton et al., 2022) or alternatively being dismissive or antagonistic (Grezes et al., 2013). Longer gaps may reflect many different cognitive processes, such as not paying attention (Bögels et al., 2015), needing to take time to respond to a difficult question (Kendrick & Torreira, 2015), conveying high-quality listening (Itzchakov et al., 2018; Van Zant et al., 2025), or savoring a shared memory together (E. M. Templeton et al., 2023).

Language

In addition to quantifying *when* people are speaking, it is also possible to analyze *what* they are saying (Yeomans et al., 2023). Doing so first requires generating a *transcript* of the conversation. Speech-to-text tools can take an audio file and transform it into text (e.g. (Radford et al., 2022)). These automated transcripts often need to be "cleaned" after the fact to correct errors. Alternatively, audio recordings can be human transcribed.

It is common practice to *preprocess* text in a transcript before applying text analysis techniques. For example, *stopwords*—articles, prepositions, and conjunctions (e.g., "the", "a")—are typically removed because they are assumed not to carry meaningful semantic content. Additional preprocessing steps can include converting text to lowercase, removing punctuation, and applying *stemming* to reduce words to their base form (e.g., "smiling" → "smile"). *Backchannels*—brief continuers that signal understanding (e.g., "yeah…")—are also common in natural conversation. Whether to exclude them depends on the research goals. For instance, if the goal is to identify conversational topics, backchannels may simply add noise. However, if the focus is on rapport building or communicative success, the presence, timing, and frequency of these turns may be crucial to analyze directly (Bergey & DeDeo, 2024; Clark et al., 2025).

Words in a transcript can be quantified in many ways. *Word frequency* provides a simple count of each unique word used. Other count-based metrics include *part-of-speech tagging*, which identifies how often words serving particular grammatical functions (e.g., nouns, verbs, adjectives) appear, and *named entity recognition*, which detects proper nouns referring to people, places, or organizations. *Sentiment analysis* estimates the overall emotional valence of words—how positive or negative the language tends to be. Dictionary-based approaches group words into predefined categories. For example, the widely used Linguistic Inquiry and Word Count (LIWC; (Pennebaker et al., 2001)) classifies words into categories including social processes (e.g., Friends, Family), affective processes (e.g., Anger, Anxiety), and cognitive processes (e.g., Insight, Inhibition).

Newer methods go beyond individual word choices to capture the underlying semantics of what people are saying within a specific context. Many of these models are trained on written text, not spoken conversational text which can be a limitation of applying them to conversation datasets.

Topic modeling can reveal the themes that emerge in conversation (Blei & Lafferty, 2009). These techniques typically require fairly large document sizes (i.e., the amount of text that gets categorized at a time) and therefore may be more appropriate at the level of an entire conversation or several minutes of a conversation at a time. Output from topic-modeling approaches are interpretable, like the dictionary-based methods described earlier. This means that a value in a particular column maps onto a particular topic. Using these models often require making certain decisions, like the number of topics which could impact the granularity of topics that get identified.

Semantic embeddings from large language models are particularly useful in mapping shorter amounts of text (e.g., at the level of a turn, sentence, or even a word) into a high dimensional feature space (Devlin et al., 2019). While the specific axes of this space are typically uninterpretable on their own, the distance between text in this space can indicate the degree of semantic similarity in meaning. Interpretation may require additional analyses, which can often be aided by the simpler lexical-based methods mentioned above, or by dimensionality reduction techniques (McInnes et al., 2018).

Audio

Beyond *what* people say is *how* they say it. Analyzing audio files can give insight into what people are doing with their voices over the course of their conversation. A wide array of acoustic features can be extracted using existing software packages (e.g.,(Boersma & Weenink, 2018; Eyben et al., 2010)). To obtain these values separately for each speaker, the voices must first be isolated—a process known as *speaker diarization*. This can be done after recording, but careful setup during data collection can greatly improve accuracy (Yeomans et al., 2023). For example, using individual lapel microphones, maintaining consistent microphone placement, and recording in acoustically dampened spaces can make diarization more reliable.

Commonly extracted acoustic features include *amplitude* (the loudness of the voice), *pitch* (the perceived highness or lowness), and *prosody* (rhythmic and intonational patterns; (Matalon et al., 2025)). More complex measures, such as *speech accommodation*, quantify the extent to which speakers' acoustic properties converge or diverge over time.

Facial movement

People also convey a lot of information with their faces during conversation (Holler, 2025). Facial analysis software typically requires being able to locate both eyes on the face, so

the ability to conduct these analyses will depend on the video recordings themselves (Cheong et al., 2023; Hu et al., 2025). Video conferencing style conversations privilege head-on face recordings and can be well-suited for these kinds of analyses. Face-to-face conversations can use 360-degree cameras on a table to capture all faces at once. Alternatively, multiple cameras can be used to capture each face individually, though they need to be synchronized in time (Cheong et al., 2019).

The general steps in a facial expression analysis pipeline include *face detection* (where a bounding box is drawn around each face), *facial landmark detection* (which allows the position of different parts of the face to be tracked), *facial identity detection* (which allows for individual faces to be linked over time), *head orientation estimation* (which details where a head sits in 3D space), *action units detection* (which estimates how much a pre-defined set of facial muscles are being activated), and *emotion expression detection* (how those combinations of action units are displaying a given facial expression). It is also possible to estimate where people are directing their *eye gaze*.

Most face expression models are trained on static images—not video with motion—and require videos to be split into static frames. Unfortunately, this means the dynamics of how facial expressions change over time are not well captured in models to date, nor how expressions change when people are actually conversing. For example, action unit 12, which indicates the degree of lip corner puller may confuse speaking with smiling behavior. In addition to problems with occlusion and head rotations, these models have been shown to perform differently based on skin pigmentation, a result of non-representative training data (Buolamwini & Gebru, 2018).

Body movement

People do not only communicate with their voices, they also communicate with their bodies. Similar to facial expression analyses, the general process for analyzing body movement requires identifying individual people in the recording, drawing a bounding box around them, and tracking their identity frame-by-frame. The ability to quantify this body movement depends on being able to see the body in the video recording. Depending on how people are positioned throughout the conversation, some parts of their bodies may be obscured by furniture in the room or by other people as they move throughout the space. Further, many of these techniques assume a static camera angle. When using multiple cameras or when needing to project features from 2D video into 3D space (to track movement, for example), transformations are needed from the field of view of the camera into "real space". The true coordinates of the room do not necessarily map onto the bounds of a field of view of an image, especially if the camera is mounted from above looking down at a space.

Motion Energy Analysis is a straightforward measure of the overall amount of pixel change from one frame to the next (Ramseyer, 2020). This captures the overall quantity of movement, though it does not differentiate between types of movement. For example, a small but meaningful head nod may generate fewer pixel changes than a less informative leg shift, simply because the leg occupies more of the visual frame.

More detailed *pose-estimation* methods identify specific body landmarks (e.g., shoulders, elbows, wrists) and track how these points move in relation to one another over time (Cao et al., 2021). When comparing poses across individuals—for example, to examine mimicry or alignment—it may be necessary to map each person into a common coordinate space. In conversations that occur while people are standing or freely moving, these methods can

additionally track where participants move, who they approach, how long they remain near one another, and how their bodies orient in relation to others (Jackson et al., 2017).

Body-movement data can also be used to study *gestures*. Automated gesture classification remains an active area of research, with accuracy depending on recording quality and the expressiveness of the behaviors.

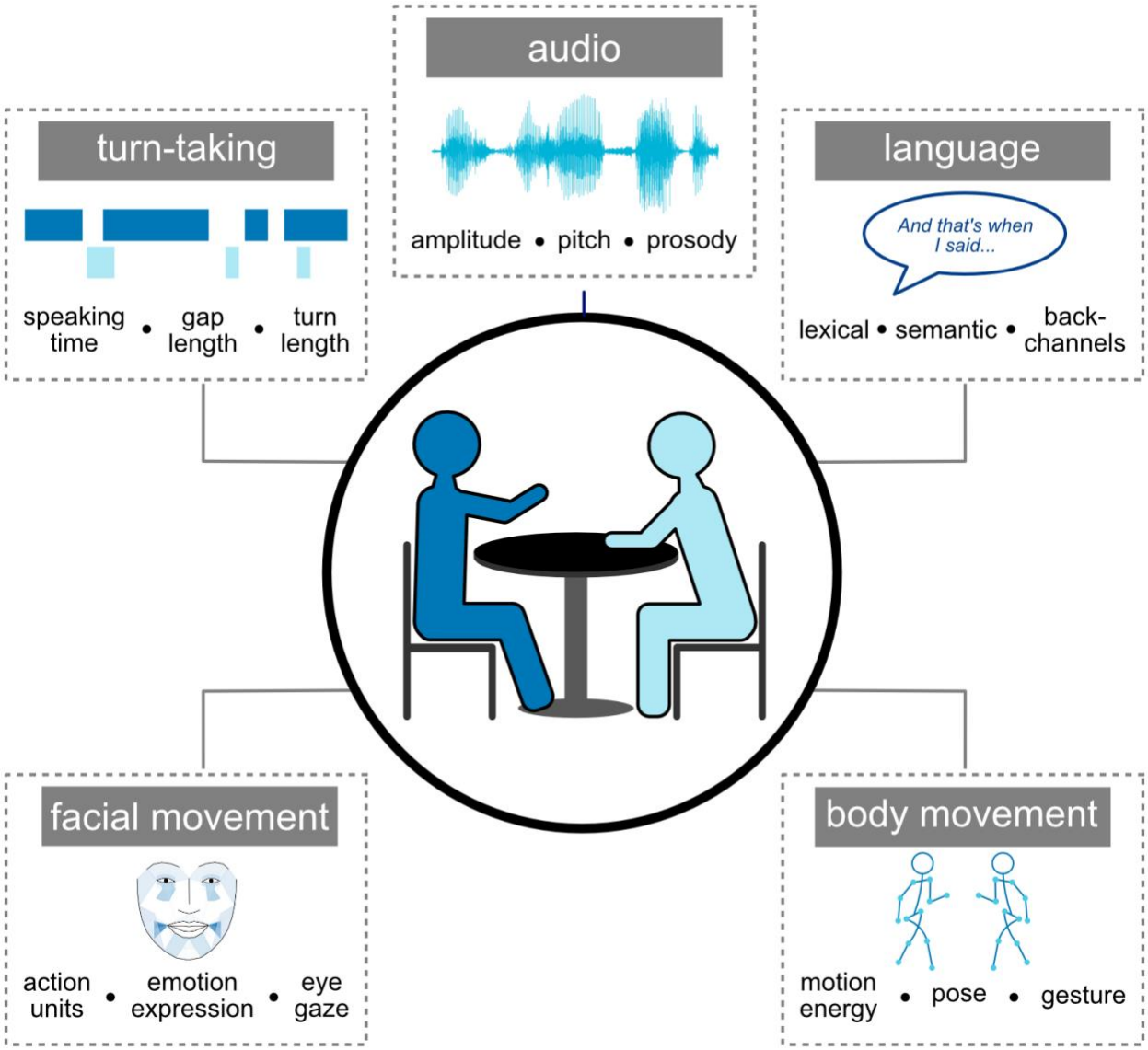

Fig 1. Feature extraction. Conversation recordings are a rich source of information that can be treated as data. Features can be extracted from video, audio, and transcriptions to quantify

different aspects of behavior. Each box describes a specific conversation modality and provides three example features that can be extracted.

Overview

In this section, we outlined several ways to extract information from transcripts, audio, and video—each capturing different aspects of conversational behavior. Quantifying these observable features provides a window into the underlying psychological processes that shape, and are shaped by, interaction. Extracting such features is an important step toward understanding how what people *do* in conversation relates to what they *think* and *feel*.

A major limitation of current feature extraction tools is that they are often not trained on natural conversational data. Collecting more of these datasets will not only deepen our understanding of conversation, but will also feed forward into the development of more specialized models—ultimately accelerating our ability to analyze conversational behavior.

Ensuring validity when using automated feature extraction

The features extracted from conversation recordings form the building blocks of subsequent analyses and are the foundation for the claims that follow. Before conducting those analyses, it is essential to ensure that these building blocks rest on solid ground. Are the feature extraction tools performing as expected on your specific dataset? Potential issues can arise both from the data input (Fig 2A) into these models and from the outputs they produce (Fig 2B).

Inspecting the inputs

A common source of error is a mismatch between what a feature extraction model expects and what your dataset actually provides. These expectations can be surprisingly specific, like a required audio sampling rate, a minimum video resolution, the visibility of both eyes for

gaze estimation, or a minimum document length for language models. Checking these details in the model documentation is critical to ensuring inputs meet these specifications.

Raw data may also contain artifacts that can produce misleading outputs. Occluded faces, abrupt lighting changes, HVAC hum in an audio recording, or transcription errors can all appear in the inputs and be misinterpreted as meaningful behavior in the outputs. Reviewing your conversation recordings can reveal artifacts in the data that would not be flagged as an issue when running the feature extraction tool. Preprocessing steps to correct these issues might include *downsampling* a high-resolution video to stabilize the frame rate, *filtering* an audio track to remove background noise, *normalizing* volume levels across speakers, or *cleaning* transcripts to fix errors.

Evaluating the outputs

Visualization. Visualization can make it easier to compare the outputs of a model with the conversation recording itself. For example, we once used a model to extract facial expressions from a video of a couple having a conversation about a recent cancer diagnosis. We inspected the output by exporting a version of the video with every frame annotated with the face location, face identity, and most likely facial expression. Reviewing this video revealed moments of mismatch. In particular, when one speaker was visibly crying, the algorithm routinely categorized her facial expression as 'happy'. This happened because the speaker had a tendency to smile through her tears, perhaps to reassure her conversation partner that she was okay. This observation immediately revealed a limitation of the model that we had used. Because it made judgements based only on the muscle movements of the face, it was unable to incorporate multi-modal information like tears. It allowed us to better appreciate that even though the output of this model might be called "likelihood of happiness" we should treat that information as "likelihood

of a smile" and nothing more. We can decide what to do with this information, based on our particular research question (Barrett et al., 2019). For example, we could quantify the presence of tears throughout the conversation to be able to examine these two features in conjunction with each other. As this example highlights, these tools are often trained on data that differ significantly from natural conversation recordings, making it essential to verify that they are working as expected (Abdurahman et al., 2024).

Inspect high vs low scores. One way to evaluate the outputs of a feature extraction model is to spot-check examples along a continuum. For example, when using a sentiment analysis model, you might sort observations by sentiment score and review the 10 most positive and 10 most negative turns. Examining these extremes can reveal whether the model's outputs make sense.

Once, when applying a dictionary-based tool to a conversation dataset, we noticed many turns were scoring high on the topic "violence". Upon inspection, we realized that topic assignment was triggered by the frequent use of the word "mean" which almost always occurred in the context of someone saying "I mean" at the start of their sentence. Taking these outputs at face value would have led us to falsely conclude that people in our dataset spent much of their time discussing violence, when they were merely using a filler phrase. This example raised immediate flags and was straight-forward to inspect. Still, something just as insidious could be happening in ways that are less obvious and far more difficult to interpret, especially when using newer text-analysis methods.

The further your analytical pipeline takes you away from the original conversation data itself, the more crucial it is to ground yourself in the conversation recording you are attempting to quantify. To give another example, we have found that what looks like large "semantic leaps"

between conversational turns can be explained by the presence of backchannel turns (e.g., "yeah…"), which tend to be semantically distinct from surrounding turns (but semantically similar to each other). Removing these turns can completely change the results, and their interpretation.

Set up tests with known labels. Another approach is to set up sanity checks that mimic the kind of data typically encountered, with clear expected outcomes. This is akin to a "unit test" commonly used in computer programming, or "eval" used to evaluate machine learning models. To continue with our sentiment example, rather than relying on the words that participants naturally use in their conversations it is possible to construct sentences that are expected to score highly positive, highly negative, or neutral and apply the sentiment package to those exemplar sentences. Does the output align with your expectations? These sorts of systematic tests can help hone your intuition about how a particular model is working and what its bounds might be.

Compare output to human ratings. A more formal evaluation would be to quantify the extent to which outputs from your feature extraction model align with human ratings. This might look like taking a subset of turns that have different sentiment scores and asking a group of participants to rate how negative or positive each turn is. You could then compute the similarity between human judgements and model output to quantify the amount of agreement between the two, similar to inter-rater reliability between annotators.

Results on these performance metrics do not need to be perfect for a model to be useful. Doing this work can help better contextualize your results or establish ceiling effects by which to compare your output. Having a large number of observations in your dataset might make your tolerance for error more forgiving whereas needing a lot of precision might encourage you to use a different model or favor human annotations over automated tools.

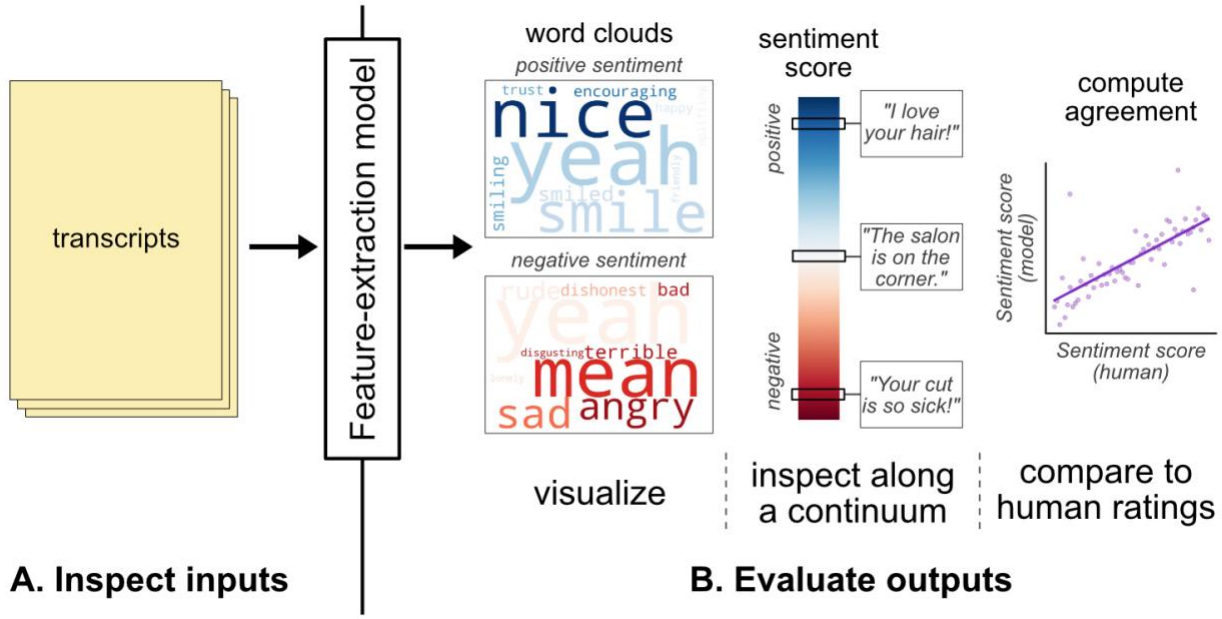

Fig 2. Ensure validity when using automated feature extraction tools. It is critical to inspect both the inputs to a feature extraction model and the outputs the model produces. To illustrate, consider a model that analyzes the sentiment (e.g., positive or negative valence) of words in a transcript. (A) Here, the inputs are the transcripts themselves. Inspecting may involve checking for missing or incorrect words, ensuring consistent formatting, or removing irrelevant information (e.g., speaker IDs or timestamps). (B) Evaluating outputs can take multiple forms. For instance, visualizing word clouds for positive and negative terms might reveal the need for additional preprocessing of the inputs, such as removing backchannels (e.g., "yeah") or applying stemming so that words with the same root are counted together (e.g., "smile", "smiled", "smiling"). Inspecting examples along a continuum can help spot-check whether the model behaves as expected and can help identify boundary conditions. For example, the slang phrase "Your cut is so sick!" likely conveys positive sentiment and is misclassified as negative. Finally, this evaluation process can be formalized by comparing model-generated sentiment scores with human-generated scores.

Overview

Preprocessing and validation steps are essential to report. They offer insight into how the model behaves on your dataset. What you learn might temper the strength of your claims, prompt you to choose a different model, combine multiple features for context, or even develop a custom model better suited to your application.

Methodological considerations

Once your feature extraction process is working as desired, those features can be used to test hypotheses and make inferences. Traditional statistical methods were developed for controlled experiments, and so natural conversation data present a very different challenge. Events occur at irregular intervals, participants contribute unevenly, and multiple behaviors unfold simultaneously. These characteristics often violate the assumptions underlying many standard parametric tests, such as independence, normality, or equality of variance. This necessitates a reliance on non-parametric methods that do not require making such strong assumptions. Here we discuss some important considerations for preparing your data for analyses (Fig 3A) and for interrogating the results of those analyses (Fig 3B).

Preparing for analyses

Selecting an appropriate unit of analysis. Different research questions call for different units of analysis. For example, if your goal is to examine the overall amount of positive language, you might compute a single positivity score for each conversation. Each row in your dataframe would therefore represent one conversation. If instead you want to capture how positivity fluctuates over time within a conversation, you might compute a positivity score for each conversational turn or for fixed time bins. In that case, each row would correspond to a turn or a time bin.

It is also important to consider whether these units of analysis occur at fixed or variable intervals. The choice between turns and time bins illustrates this distinction. Turns vary in length, so turn-level data occur at irregular intervals. Time bins provide regular spacing but can introduce other peculiarities. For example, speaker switches rarely align perfectly with bin boundaries, which means a single bin may contain contributions from multiple speakers in

unequal proportions. Neither approach is inherently right or wrong—the appropriate unit depends on the phenomenon you aim to capture.

Normalization. Different research questions will call for different types of data normalization, and these decisions can meaningfully alter results. When trying to make comparisons across conversations that are different lengths, it is important to consider them on the same scale. After all, longer conversations have more opportunities for specific events to happen. One strategy can be to normalize timepoints by the total conversation length, transforming timepoints into the percent of the total conversation.

Oftentimes, you may be more interested in how participants deviate from their own baseline behaviors rather than simply looking at absolute differences in behaviors, especially when individual participants behave in systematically different ways. For example, one person might generally emote more intensely than another, but you may be interested in examining if their emotion dynamics change at the same timepoints. In this case, normalizing within each participant (e.g., by z-scoring or mean-centering) can be useful.

Other times, you may want to consider multiple features in a single model (Hyekang Joo, 2025). If those features are expressed on vastly different scales, you may want to normalize each of them so that they have the chance to exert equal influence in the model.

Accounting for dependencies. Natural conversation data contain dependencies at multiple levels, and these must be accounted for in statistical models. A useful starting point is to examine the structure of your dataframe: which pieces of information repeat across multiple rows? These repeated elements (e.g., participant IDs, dyad IDs, conversation IDs) often represent the levels you want to generalize beyond. For instance, researchers typically aim to make claims about people in general, not only the individuals in their datasets. Proper modeling helps ensure

that inferences reflect these intended generalizations (del Rosario & West, 2025; Westfall et al., 2016; Yarkoni, 2020). Random intercepts can account for systematic differences between participants or dyads, while random slopes (e.g., for time) allow individuals or dyads to vary in how these differences unfold (Kenny et al., 2021).

When examining how features change over the course of a conversation, it is also important to consider the role of *autocorrelation*, where adjacent timepoints might necessarily be more similar to each other. It may be appropriate to remove these effects by linearly detrending the data before entering it into models or by including time as a covariate.

Interrogating analyses

The details of your specific analysis will be highly dependent on your research question. Here, we discuss some aspects of interpreting analyses that are generally applicable when working with conversation data.

Establishing statistical significance. For the majority of studies in psychology, establishing significance is straightforward and baked into the experimental design (e.g., comparing two conditions with a t-test, with the null hypothesis that the groups do not differ). However, conversational data pose a different challenge. Within a single conversation, many behavioral signals covary with each other, with that redundancy helping facilitate clear communication (Botch & Finn, 2025; Degen et al., 2020). As a result, simply observing a relationship between two features is not enough to conclude that a meaningful effect has occurred. The key question becomes whether an observed effect exceeds what would naturally arise from the inherent structure of conversational data.

To answer this, it is important to construct null distributions to represent what the data would look like in the absence of the specific effect being tested. The choice of null depends on

the claim you would like to evaluate. For dyad-specific effects (e.g., whether two people align more than would be expected from any two strangers), you might create pseudo-pairs by recombining participants who never interacted. To test for temporally specific effects (e.g., whether synchrony depends on precise timing), you might use a surrogate data analysis strategy like circular time-shifting or phase randomization, which preserve overall temporal structure while disrupting alignment (L. Chang et al., 2020; Lancaster et al., 2018). Comparing an observed effect to these null distributions tests whether the observed effect is larger than what would be expected by chance given the data's natural dependencies.

Evaluating robustness to analytical decisions. Analyzing natural conversation data often requires making decisions that do not have a single "correct" answer. Choices such as how to preprocess text, the number of clusters in unsupervised clustering methods, or the timescale for binning data can all influence downstream results (Gautheron et al., 2025). Rather than seeking a single "best" approach, researchers should report how results change under various plausible decisions (Steegen et al., 2016). This process can reveal informative patterns. For example, if an effect disappears at certain timescales, it suggests meaningful temporal boundaries. If different text preprocessing steps alter results, it suggests key linguistic features driving the effect. The goal is not for every effect to remain unchanged across all decisions (Auspurg, 2025), but to use any systematic differences as useful information about the construct itself. Understanding (and reporting) these variations can generate new hypotheses, clarify the limits of an effect, and help readers evaluate where and how it may generalize across different datasets.

Following up on initial findings. After establishing a relationship between a conversation feature and an outcome in your dataset, (and convincing yourself that it is robust

and significant, based on the points above), there are many ways to think about following up on that finding. One way is to try and establish the causal nature of that relationship. This can be done using the same dataset, for example with methods that establish the temporal precedence between two time series (Epskamp et al., 2018; Granger, 1969; E. Templeton et al., n.d.). Another approach is to run a targeted follow-up experimental study where intervening on a feature should have a predictable effect on a given outcome (E. M. Templeton et al., 2022).

Next, it is often important to explore the mechanism behind a given relationship. *Why* does a change in X have a corresponding change in Y? Computational modeling can be a useful tool here, to help explore the impact of possible latent constructs, especially when combined with targeted experimental studies that explicitly measure these candidate mechanisms.

Finally, it is essential to test whether this observed effect generalizes to other conversational contexts. Does this finding represent a "universal process" or are these effects more specific? Researchers can investigate myriad roles of actor, partner, group size, goals, culture, development, and communication modality in potentially moderating these effects.

No single study can fully capture the complexity of conversation. Progress in this field will depend on integrating findings across many types of conversations, extracting different sets of features, and modeling their relationships to psychological outcomes using a range of methodologies, including self-report (O'Nell et al., 2025; E. M. Templeton et al., 2022), behavior (Di Stasi et al., 2023), causal manipulation (Arias-Sarah et al., 2024; Mahaphanit & Chang, 2023), and computational modeling (Doyle & Frank, 2016; Hawkins et al., 2017; Qian et al., 2024). When a finding emerges consistently across different methods and datasets, confidence in that finding increases (Abney et al., 2014). Over time, this approach will build a

foundational understanding of conversation, revealing generalizable patterns, underlying mechanisms, and predictive models.

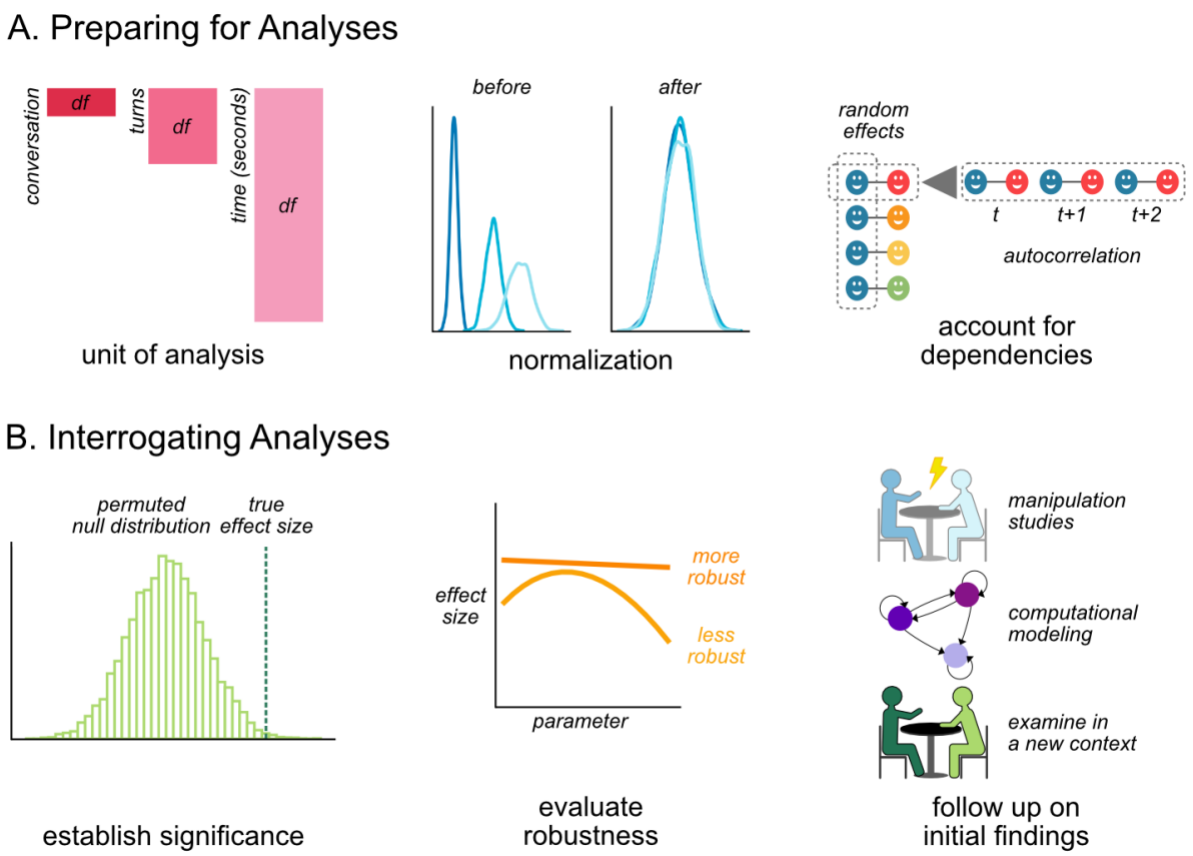

Fig 3. Methodological considerations. (A) When preparing data for analyses, it is important to (i) consider the unit of analysis (e.g., what each "row" of a dataframe will represent), assess whether normalization is needed (e.g., for feature scaling or comparing conversations of different lengths), and (iii) account for dependencies in the data (e.g., subject-level random effects or temporal autocorrelation). (B) When interrogating analyses, it is often appropriate to (i) compare effect sizes against null distributions generated by permutation tests, (ii) evaluate how any effects of interest vary as a function of arbitrary decisions in an analysis pipeline, and (iii) follow-up results with targeted experimental studies, computational modeling, or testing generalization to a new context.

Conclusion

Being able to study conversation should be of interest to all scientists, not only those who primarily study social processes. Afterall, conversation is the outward expression of our inner states. Studying what people say to each other provides a more direct window into their thoughts

and feelings than a number on a Likert scale or performance on a laboratory task. Developing the pipelines, methods, and tools to study conversation will enrich every area of psychology, opening unexpected directions for discovery.

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